Predictive Analytics

Lecture 4

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Chapter 7:
Time Series Regression Models

Predictors

- **A predictor variable** is an explanatory variable (also called independent variable or a regressor)
- Our outcome variable (what we are forecasting) is called the forecast variable or the dependent variable (sometimes also called the explained variable or the regressand)

Predictors

- We have already seen in earlier lectures that many time series have trends, seasonality and/or persistence
- We can include **predictors** for these patterns in our forecasting model

Some useful predictors in time series analysis

- Dummy variables
- Seasonal dummy variables
- Trend variable
- Lags

Dummy variables

- Dummy variables are simply variables that take value 1 if some statement is true, and 0 otherwise

- E.g. lets say you have daily ice-cream sales as your outcome variable: you can include variable D where D = 1 if it was sunny during the day and D = 0 if it was not
- Usually we want to name our dummy variables by the "true-statement," in this case I would name the variable *sun*
- If a variable is named *female* it suggests it takes value 1 if the observation refers to a female, 0 otherwise
- If a variable is named *male* it suggests it takes value 1 if the observation refers to a male, 0 otherwise

Seasonal dummy variables

Note that in the table you have dummy variables for each weekday, except for Sunday

Why would we not include seven predictor variables for the seven days of the week?

	$d_{1,t}$	$d_{2,t}$	$d_{3,t}$	$d_{4,t}$	$d_{5,t}$	$d_{6,t}$
Monday	1	0	0	0	0	0
Tuesday	0	1	0	0	0	0
Wednesday	0	0	1	0	0	0
Thursday	0	0	0	1	0	0
Friday	0	0	0	0	1	0
Saturday	0	0	0	0	0	1
Sunday	0	0	0	0	0	0
Monday	1	0	0	0	0	0
⋮	⋮	⋮	⋮	⋮	⋮	⋮

Trend predictor

- It is common to use trend as an predictor when we are working with time series. I.e. we use t = 1, …., T as a control/predictor

$$y_t = \beta_0 + \beta_1 t + \varepsilon_t,$$

Lags

- In many cases we want to include lagged values of our predictors
- E.g. if I want to look at money spent on advertising as a predictor of the number of tourists in Iceland, then advertising in previous months may be more relevant than advertising today

$$
\begin{aligned}
x_1 &= \text{advertising for previous month;} \\
x_2 &= \text{advertising for two months previously;} \\
&\vdots \\
x_m &= \text{advertising for } m \text{ months previously.}
\end{aligned}
$$

What if we have more useful information for our forecast?

The future depends on the past

- Until now we have focused on models where the previous values of a time series predict its future
- $\hat{y}_{t|t-1} = f(y_{t-1}, y_{t-2}, \dots)$
- E.g. Naïve model, mean model, drift model…
- Today: What if **other variables** contain useful information about the future

Are we leaving information on the table

- Suppose inflation depends not only on past inflation…but also on past unemployment, oil prices, exchange rates, wages…
- How do we make use of that information?

Including other variables in our forecast

- Do the two variables move together?
- Or perhaps in opposite directions, i.e., when one increases the other decreases?
- Does the new variable add information?
- Is the data available at forecast time?
- Does the relationship between the two variables make economic sense?

Example: Consumption and Income

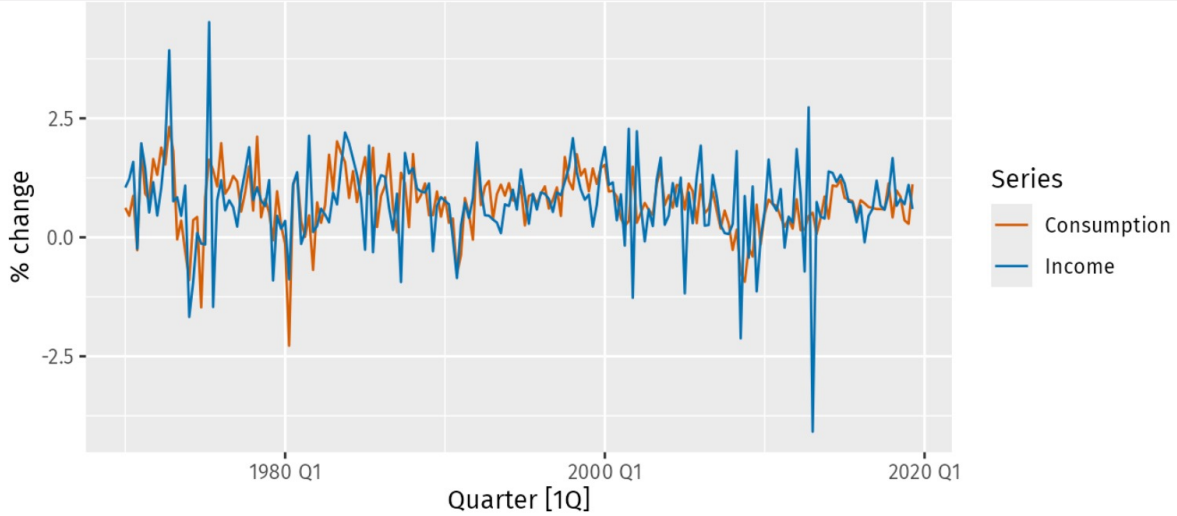
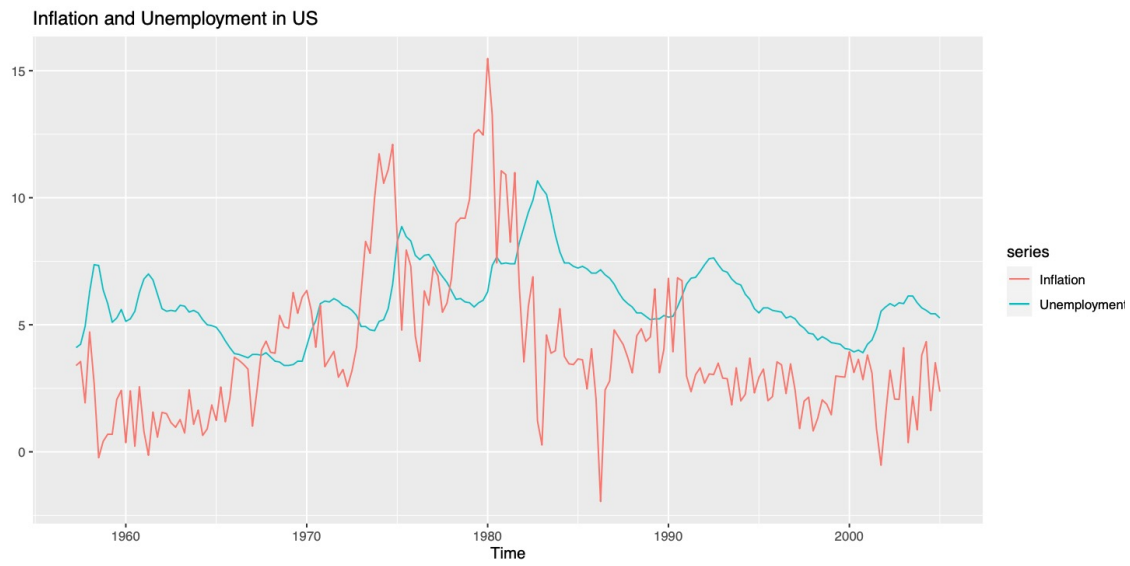


Figure 7.2: Percentage changes in personal consumption expenditure and personal income for the US.

Let's check the relation between inflation and unemployment



Another example: inflation and unemployment

How can we make use of these co-movements?

- If two variables move together over time, we can write
  - Outcome = systematic part + unpredictable part
- For example: $\text{Inflation}_t = \text{Unemployment}_t + \text{error}$

We can expand our forecasting model further

- E.g.
- $\text{Inflation}_t = \text{Inflation}_{t-1} + \text{Unemployment}_t + \text{Exchange rate}_t + \text{error}$
- Note: A regression model can describe relationships at time t. But a forecasting model must respect the information available at the time the forecast is made.

Time series regression model

- The time series regression model allows us to think about the relationship of our time series *y*, with another time series *x*
- Instead of relying solely on past values of *y* in our forecast we can now include other sources of information
- E.g., *y* is monthly ice-cream sales, and *x* is average monthly temperature. *x* might allow us to get a more complete picture and thereby enable us to more accureately forecast the future sales

Time series regression model: linear case

In the simplest case, we assume a linear relationship between the forecast variable *y* and a single predictor variable *x*

$$y_t = \beta_0 + \beta_1 x_t + \varepsilon_t$$

$y_t = \beta_0 + \beta_1 x_t$ is the explained part of the model

ε_t is the error term or the deviation from the underlying model

ε_t captures anything that may affect y_t other than x_t

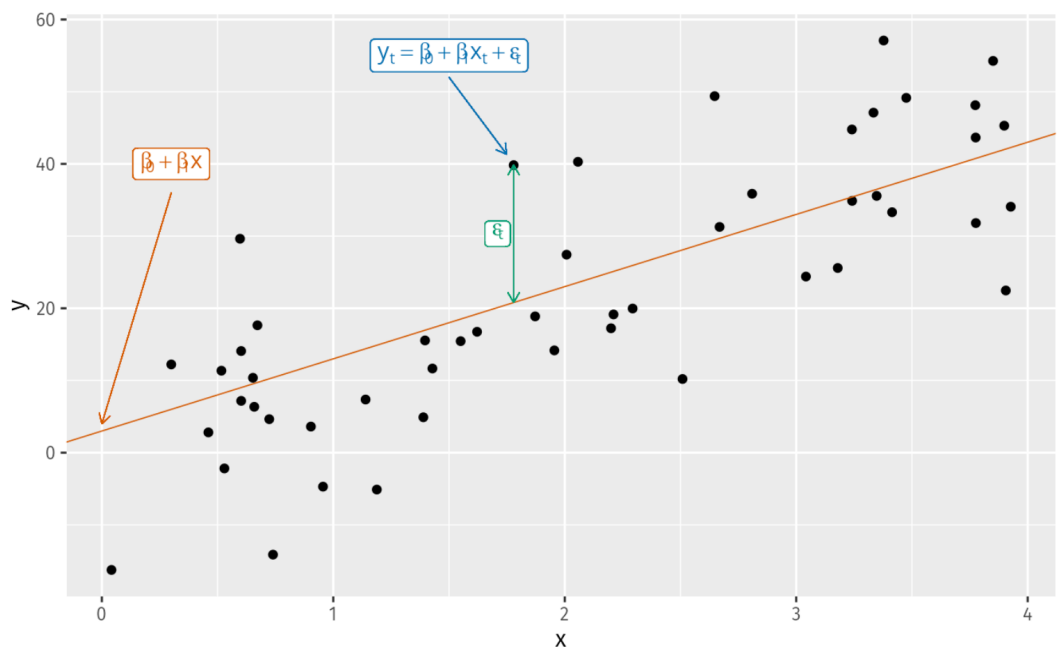


Figure 7.1: An example of data from a simple linear regression model.

How are the coefficients chosen?

- The Regression chooses the coefficients (β_0, β_1,…) to minimize the prediction errors in the training data
- That is, it finds the combination of predictors that best fits the past data

Time series regression model: linear case, example

$$\hat{y}_t = 0.54 + 0.27x_t$$

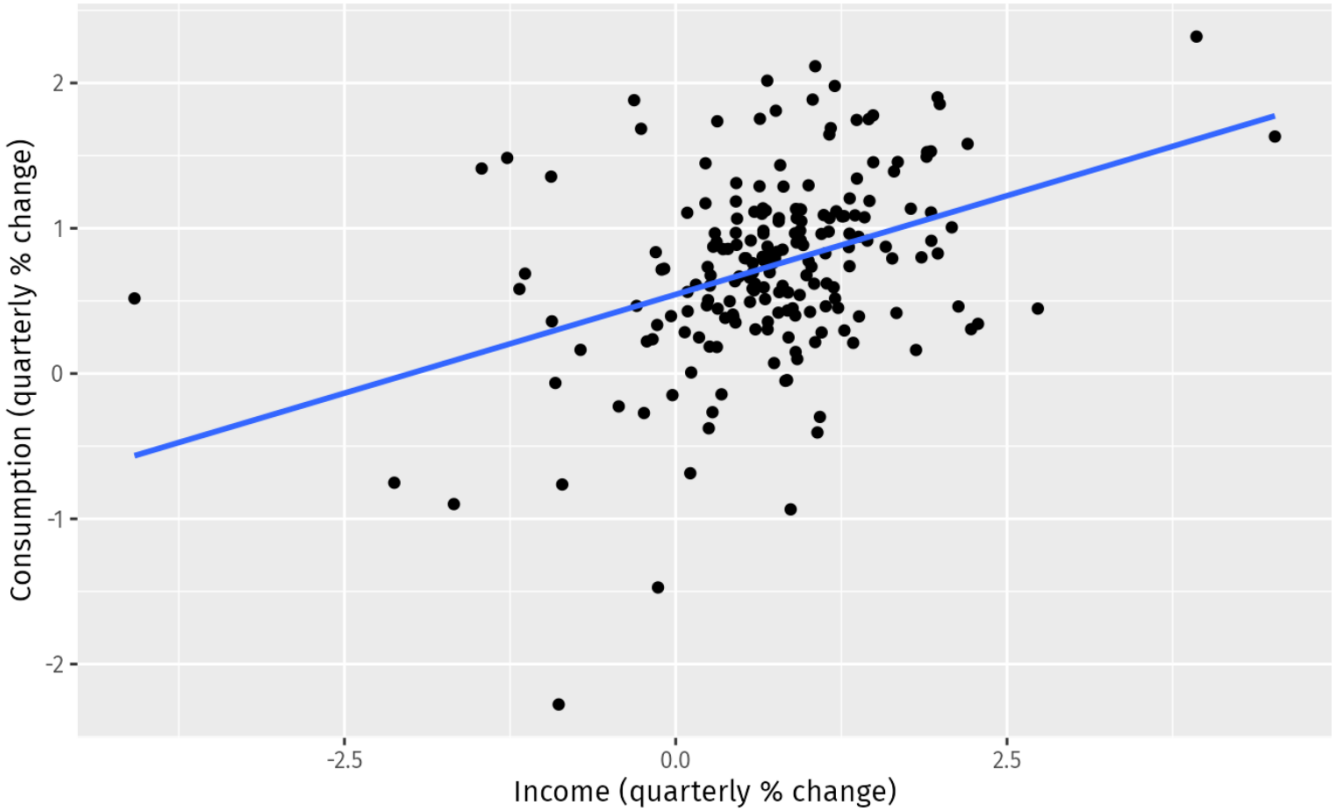
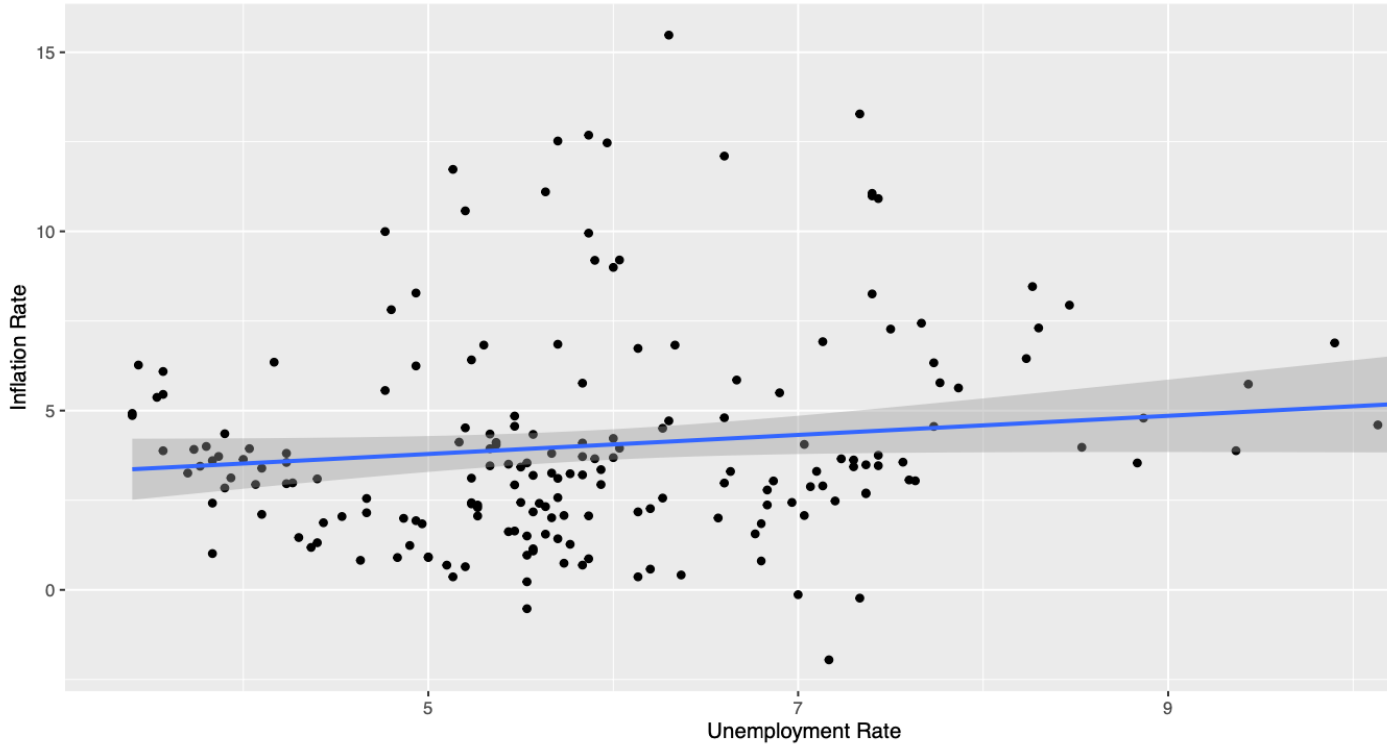


Figure 7.3: Scatterplot of quarterly changes in consumption expenditure versus quarterly changes in personal income and the fitted regression line.

ɔecify the relation $infl_t = c + \beta unemp_t + \varepsilon_t$



When we estimate this relation with the function `model(TSLM())` we get

	Model 1
Intercept	2.4593***
	(0.9101)
Unemployment	0.2661*
	(0.1499)
R^2	0.0163

Table: Regression Model (Standard Errors in Parentheses)

- Positive relationship between inflation and unemployment

Example: Inflation and Unemployment

Combining multiple time series

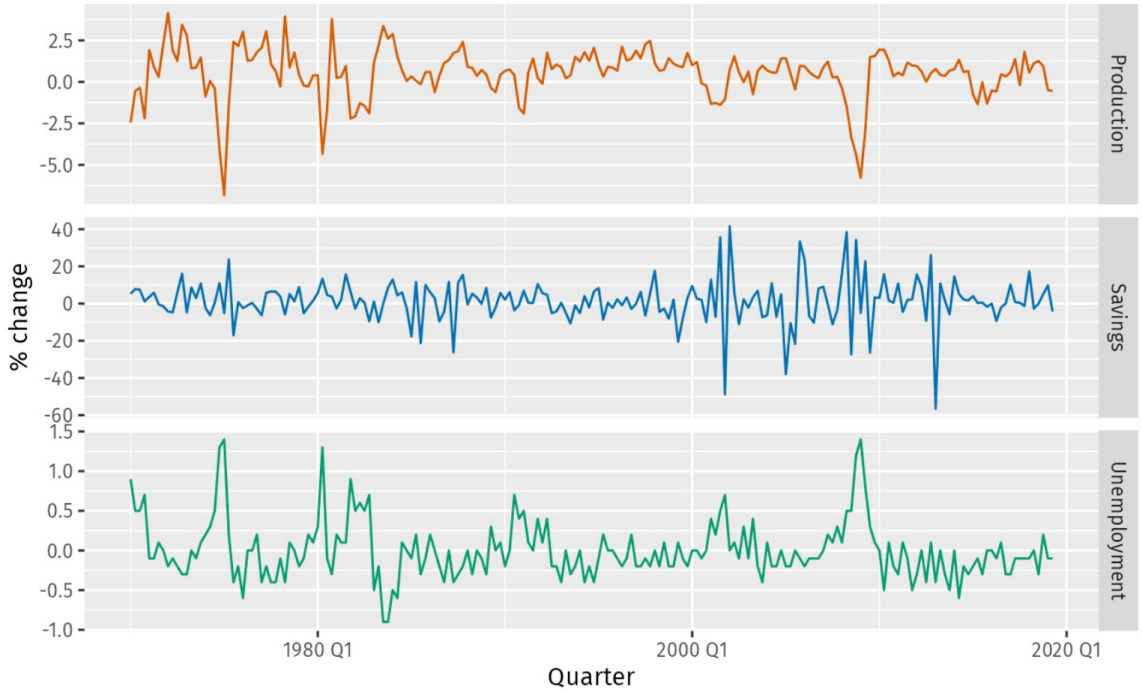


Figure 7.4: Quarterly percentage changes in industrial production and personal savings and quarterly changes in the unemployment rate for the US over the period 1970Q1-2019Q2.

Combining multiple time series

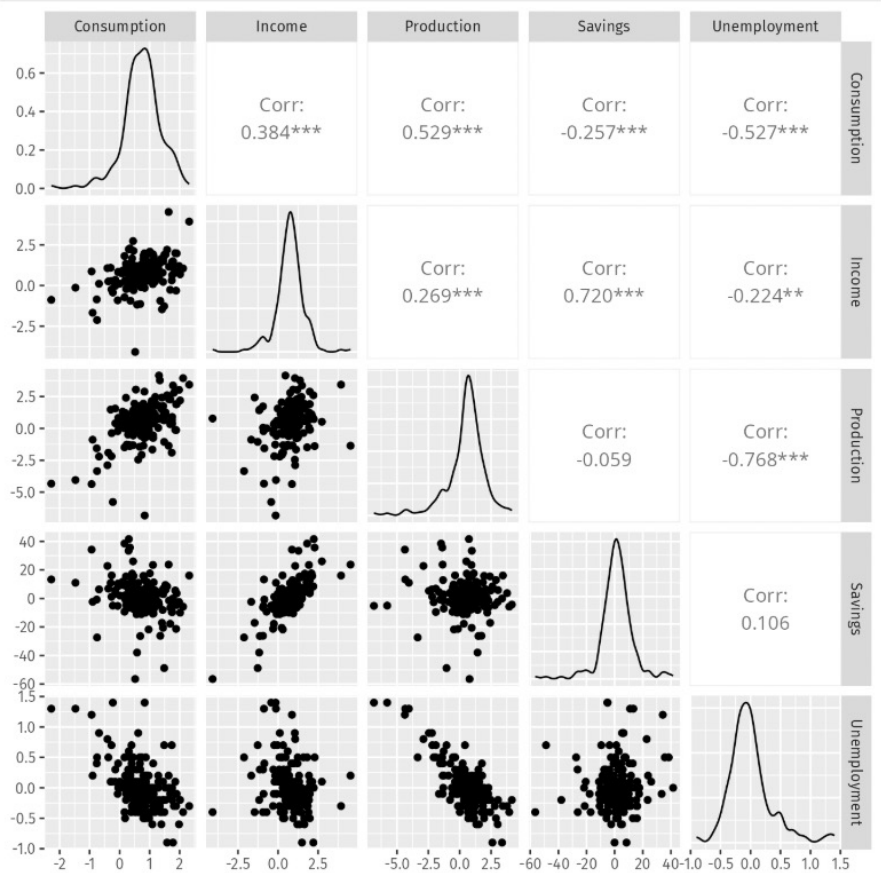


Figure 7.5: A scatterplot matrix of US consumption expenditure and the four predictors.

Multiple linear regression

- In many cases we can improve the accuracy of our model by looking at multiple predictors:
- $y_t = \beta_0 + \beta_1 x_{1,t} + \cdots + \beta_1 x_{K,t} + \varepsilon_t$
- Which can be written in a matrix notation: $y = X\beta + \varepsilon$
- E.g., before we looked at income as a predictor for consumption
- Perhaps we can increase the accuracy of our model by also looking at saving, unemployment and production

How to choose predictors?

- Try to rely on theory and intuition
- Do the predictors make sense from a theoretical perspective?
- It is not sufficient that models perform well statistically – preferably they also need to align with theory and/or knowledge
- E.g. you are interested in predicting ice-cream sales: you observe seasonality in your time series, and when you look at data for swimming pool visits you see similar seasonality – should you include swimming pool visits as a predictor?

How to choose predictors?

- Information criteria like **Akaike's Information Criterion (AIC)** or **Schwarz's Bayesian Information Criterion (BIC)** can provide statistical guidance
- Adding predictors will (weakly) increase R^2 but AIC and BIC penalise for adding too many predictors, if they do not sufficiently improve the fit
- AIC and BIC may suggest that it is a good idea to add certain predictors – but always consider whether it also makes sense from a theoretical perspective (why would there be a relationship between the predictor and the outcome variable?)

How to choose predictors?

Table 7.1: All 16 possible models for forecasting US consumption with 4 predictors.

Income	Production	Savings	Unemployment	AdjR2	CV	AIC	AICc	BIC
●	●	●	●	0.763	0.104	-456.6	-456.1	-436.9
●	●	●		0.761	0.105	-455.2	-454.9	-438.7
●		●	●	0.760	0.104	-454.4	-454.1	-437.9
●		●		0.735	0.114	-435.7	-435.5	-422.6
●	●		●	0.366	0.271	-262.3	-262.0	-245.8
	●	●	●	0.349	0.279	-257.1	-256.8	-240.7
●			●	0.345	0.276	-256.9	-256.6	-243.7
●	●			0.336	0.282	-254.2	-254.0	-241.0
	●	●		0.324	0.287	-250.7	-250.5	-237.5
		●	●	0.311	0.291	-246.9	-246.7	-233.7
	●		●	0.308	0.293	-246.1	-245.9	-232.9
	●			0.276	0.304	-238.1	-238.0	-228.2
			●	0.274	0.303	-237.4	-237.3	-227.5
●				0.143	0.356	-204.6	-204.5	-194.7
		●		0.061	0.388	-186.5	-186.4	-176.7
				0.000	0.409	-175.1	-175.0	-168.5

What if it predicts well but makes no sense?

Example

- Suppose Instagram activity predicts inflation extremely well.
- It improves your out-of-sample forecasting accuracy.
- But you don't understand why.
- Do you include it in your forecasting model?

Forecasting vs understanding

- For policy, or for research: We are usually interested in the mechanisms
- If we are purely interested in forecasting we do not necessarily care
- But if the relationship is not meaningful we should be careful…

Correlation is not causation

- Correlation $\neq$ Causation $\neq$ Forecasting
- Lets say we have two variables x and y that move closely together – it is not necessarily the case that x is causing y
  - Spurious regression
  - Reverse causality
  - Omitted variables
- **Confounder** is a variable that is not included in our forecasting model but influences both the response variable and at least on predictor variable
- **Correlation**s can be useful in forecasting, even though they do not imply **causation**
- However, understanding causation and picking up confounding factors can improve the forecast accuracy

Correlation is not causation

- It can be accepatable to use predictors that are correlated with the outcome, even if there is no causal relationship – especially if you are purely interested in making the best predictions
- However, there is an increased emphasis on understanding the underlying mechanisms driving the data
- We usually want to build models that not only predict accurately but also provide insights into e.g. how the economy works
- Knowing the mechanisms can inform both financial and policy decisions – and help us anticipate how changes in one area may affect others

Spurious Regression

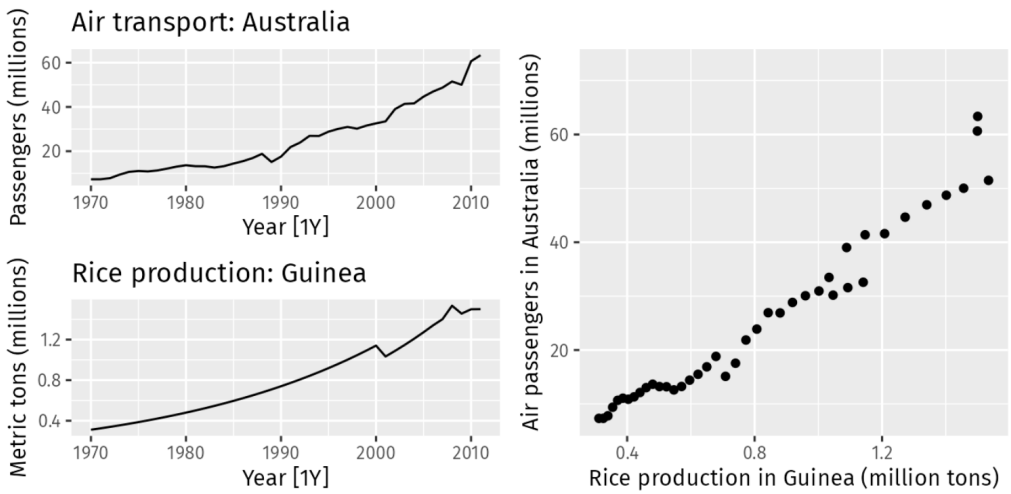


Figure 7.12: Trending time series data can appear to be related, as shown in this example where air passengers in Australia are regressed against rice production in Guinea.

Spurious regressions

- Sometimes we can have a model where the estimation suggests a highly significant relationship between *x* and y
- Results show high R^2 and significant t-statistics, but the results have no economic meaning

The risk

- If the relationship between the two variables is not structural, i.e., there is not underlying mechanism, it may easily disappear when the environment changes
- Since we do not understand the relationship between x and y, we do not know when or how the relationship will break, and we can't predict how it will react to shocks
- That is our forecasting model may work, until very suddenly it does not

The caveat

- Even structural relationships can change over time, and time series forecasting is always partly uncertain
- But understanding the underlying mechanisms improves the robustness of our forecasting models

Evaluating the regression model

Evaluation of our forecasting models

- Last week we saw how we evaluated the performance of some simple benchmarking models
- The forecasting workflow is the same as before:
- I.e. with the regression models we still follow the same steps:

1. Split data into training and test sets
2. Estimate the model using training data
3. Check whether residuals look random
4. Generate forecast for the test period
5. Look at the forecast errors (MAE, RMSE, etc.)
6. Compare models using out-of-sample performance

What is “new”?

- We now use multiple time series
- In some cases we may also need to forecast the predictors
- The risk of overfitting increases
- Interpretation may become more complex
- But hopefully our forecasts are also becoming more accurate ☺

Evaluating the regression model

- After estimating the model, we look at:

1. Do the residuals look random?
2. Is there leftover autocorrelation?
3. Does this more complicated model improve the out-of-sample forecast compared to the benchmark?

Note on overfitting

- If we add predictors R^2 will increase, as our in-sample errors will decrease
- But the out-of-sample performance may worsen!
- We just saw how we can use AIC and BIC to choose predictors
- These criteria penalize complexity – and therefore help balance fit and simplicity

Evaluating the regression model

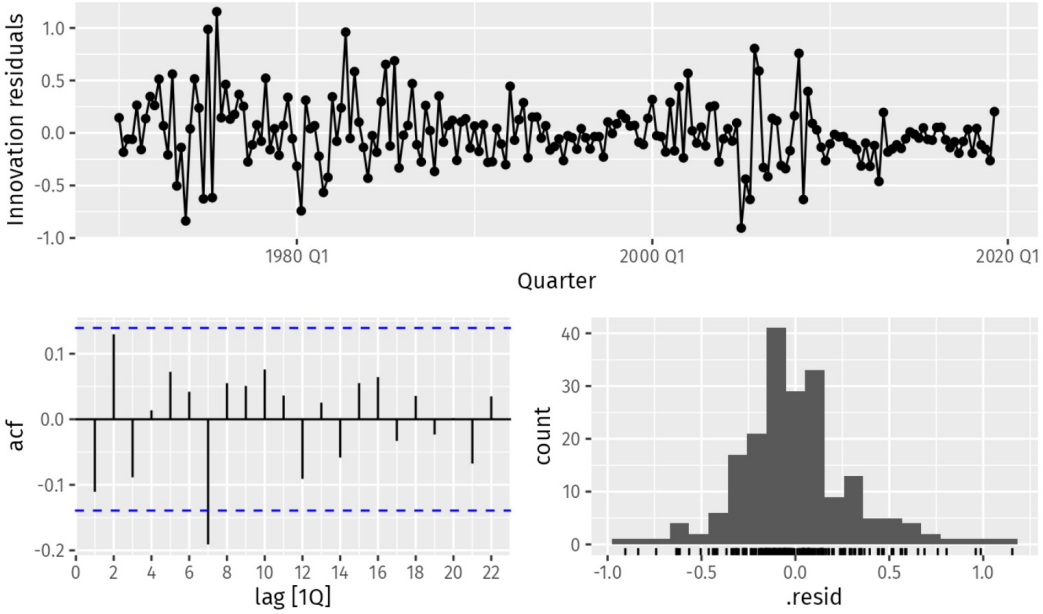


Figure 7.8: Analysing the residuals from a regression model for US quarterly consumption.

Evaluating the regression model

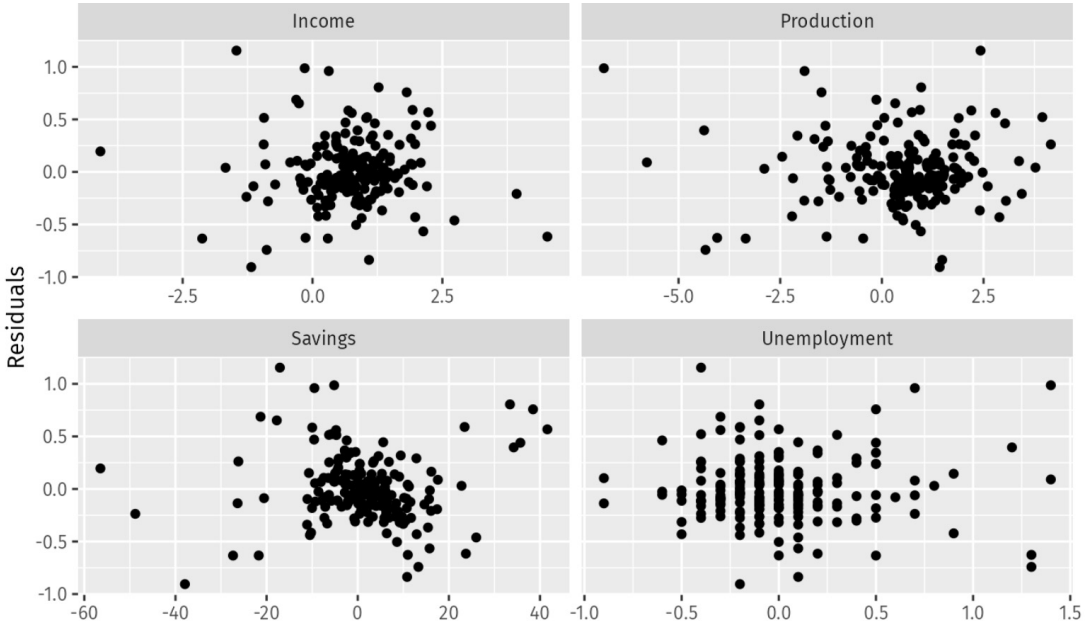


Figure 7.9: Scatterplots of residuals versus each predictor.

Forecasting with regression

Forecasting with regression

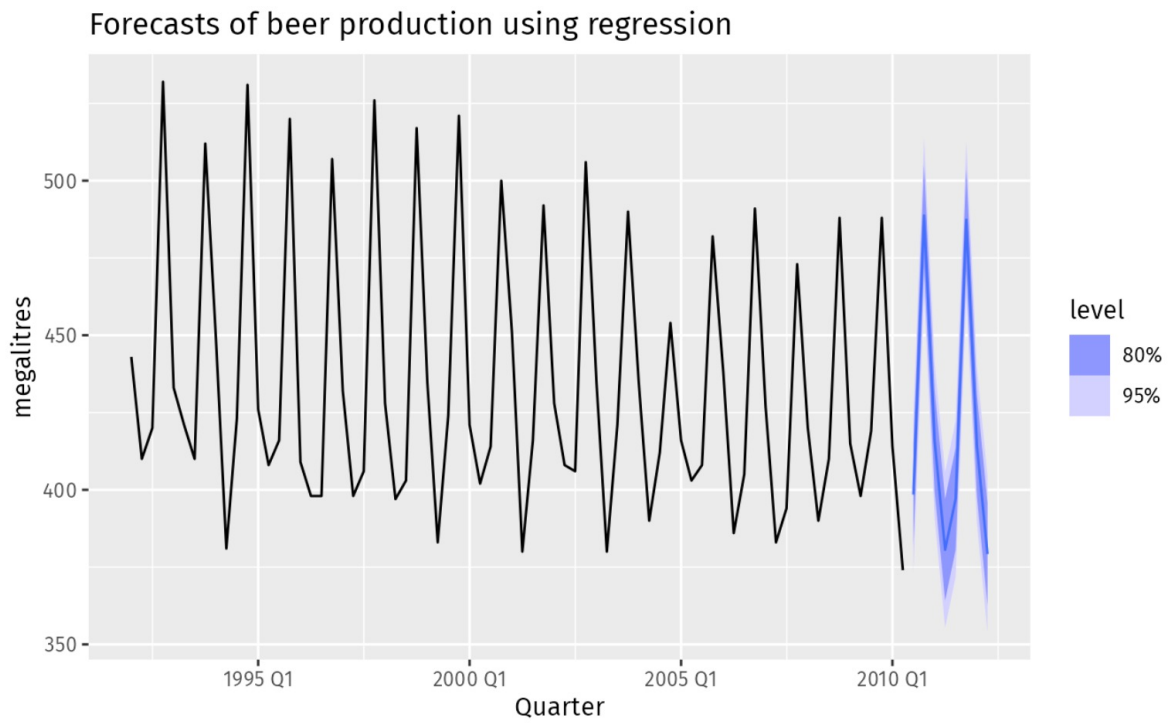


Figure 7.17: Forecasts from the regression model for beer production. The dark shaded region shows 80% prediction intervals and the light shaded region shows 95% prediction intervals.

Example: We want to forecast inflation next month

- We need coefficient estimates for our forecasting model
- We need values for the predictors
  - Us lagged values
  - Forecast predictors
  - Or assume scenarios

Forecasting with regression

- Ex-ante forecast: using only information that is available in advance. This usually means that we need to forecast the predictors. But sometimes we know the predictors in advance (e.g. trend and seasonaly dummies)

- Ex-post forecast: Use later information. These are not exactly forecast (we are forecasting something that already happened) but can be useful to understand our models

Scenario based forecasting

- Sometimes we assume values for predictors and compute the forecast:
- What happens to inflation if oil prices increase 10% vs decrease 10%
- What happens to inflation if unemployment rises 1% vs drops by 1%

Scenario based forecasting

E.g. How much will consumption change when there is 1% constant income growth, 0.5% constant savings growth, and no change in employment

Vs if there is 1% constant decline in income, 0.5% decline in savings, and no change in employment

I.e. we assume particular values for the predictors

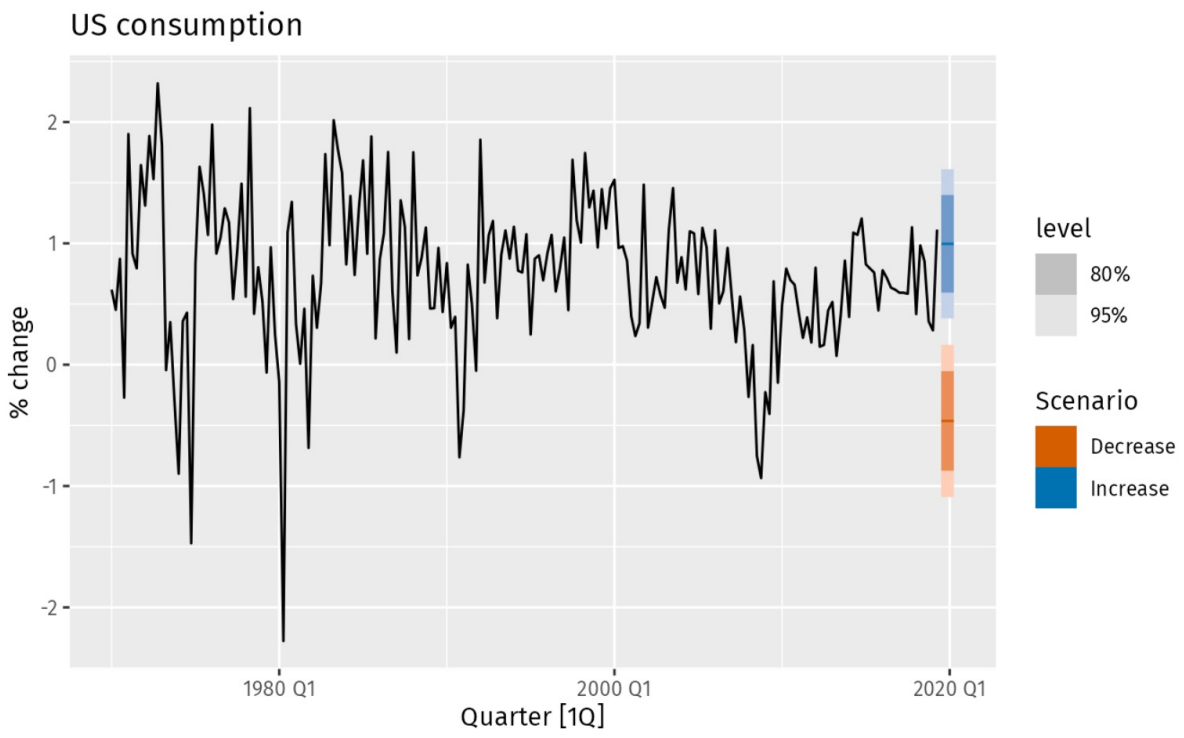
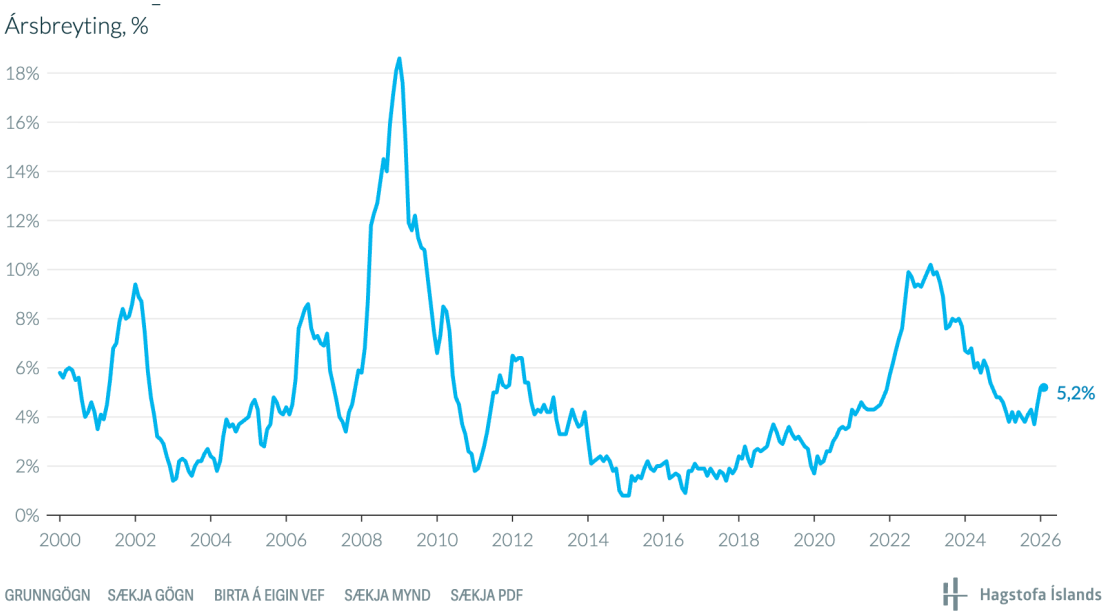


Figure 7.18: Forecasting percentage changes in personal consumption expenditure for the US under scenario based forecasting.

Exercise, predicting inflation in Iceland



Exercise, predicting inflation in Iceland

- You work at the Central Bank of Iceland and are asked to forecast inflation 1-2 years ahead
- You have data on inflation from 2000
- Your task is to build a reasonable forecasting model

Exercise, predicting inflation in Iceland

- Factors to consider:
- Does inflation show persistence?
- Is there a trend?
- Are there structural breaks and/or crisis periods?

Exercise, predicting inflation in Iceland

- You have access to the following variables:
  - Lagged inflation
  - Exchange rate
  - Oil prices
  - Wage growth
  - Unemployment
  - Crisis dummy
- Which variables are good for fitting the past, and which are good for forecasting?
- Which relationships are stable?
- Would you use a different set of variables if you were asked to forecast inflation ten years 5 years ahead?

Exercise, predicting inflation in Iceland

- What do you think are some important predictors to include? Does it make sense to include lagged values of your predictors?
- How would you try to capture the "unusual periods" e.g. the financial cricis, and the covid years?
- Do you think your model will be useful to forecast inflation in the next 1-5 years?
- Are there any variables that you would include if your objective is to fit the data, but not if you would like to forecast the future?

Summary of today

- Today we learned

1. How to expand our information set to improve our forecasts
2. How to include trend, seasonality, lags, and other time series as predictors
3. We can only use information available at prediction time – sometimes we need to forecast our predictor variables as well
4. How to think about correlation between variables when choosing predictors
5. We saw that the evaluation of regression models is the same as with our benchmark models

Next week

Chapter 8, exponential smoothing